SUPPLEMENTAL QUALITY ASSURANCE PROVISION (SQAP)

Prescribed by: eSOP 4155.05-001
Sponsor: J3

REVISIONS

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1	ENERGY QAP E1	04/11/2024	

ENERGY QAP E1 CONTRACTOR INSPECTION RESPONSIBILITIES (April 2024)

(a) The following requirements of this Quality Assurance Provision (QAP), paragraphs (a) (1) through (a) (3) and applicable tables apply to JET A/JET A-1, with or without additives, transported by fungible pipeline. These requirements also apply to terminals that receive fungible JET A/ JETA-1 via pipeline, then subsequently ship via tank truck/trailer and rail car.

(1) QUALITY CONTROL PLAN (QCP)/COMPANY PLAN (QUALITY) - The contractor shall, as a minimum, have a written QCP/ company plan in effect that ensures the quality of all aviation fuel supplied in performance of this contract. A copy of the QCP, in English, shall be presented to the Quality Assurance Representative (QAR) for their review and acceptance prior to commencement of production or services. An acceptable QCP is required prior to Government inspection and acceptance of supplies or services. The QCP shall be reviewed annually and updated when deemed necessary by the Contractor or the Government, to include changes necessary to prevent the recurrence of quality problems. The annual review shall be performed by the Contractor and shall be documented in a manner that allows the QAR to validate the review date and who performed it. The Contractor shall sign and date the original QCP, as well as each subsequent revision. Revisions shall also be presented to the QAR for review and acceptance prior to implementation. When acceptable to the Government, the original QCP, and any revisions, shall be signed and dated by the QAR. The requirements outlined in the QCP do not change requirements of the contract; any changes to contract documents are executed by the Contracting Officer. The following minimum elements are required:

(i) Contact information for key operational positions and a schematic diagram of facilities pertinent to the inspection system, indicating all inspection points.

(ii) Procedures for full specification (A-Level) testing, which shall be performed on each batch tank offered under this contract, and documentation of those test results on a Certificate of Quality (CoQ), Certificate of Analysis (CoA), or equivalent that provides the required test results.

(iii) Procedures for line injection of all additives at all locations within the Contractor's facility, to include where and when each additive will be line injected, how to ensure that all additives are homogeneously blended into the aviation fuel and that no premixing of additives occurs.

(iv) Procedures for inspecting/ensuring tank trucks/trailers, intermodal/ISO Containers and rail cars are suitable to load the aviation fuel furnished under this contract.

(v) When shipping via pipeline, procedures that ensure the aviation fuel being introduced complies with the specification in the contract and the pipeline company's specification for fungible aviation fuel.

SUPPLEMENTAL QUALITY ASSURANCE PROVISION (SQAP)			DEFENSE LOGISTICS AGENCY, ENERGY 8725 JOHN J. KINGMAN RD FORT BELVOIR, VIRGINIA 22060-6222		
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(vi) Documentation:

(A) Prior to shipment and for each shipping tank, a copy of the CoQ, CoA, or equivalent, shall be provided to the QAR/Government representative for approval prior to shipment and the results shall be entered into the Statement of Quality Tab during submission of an Energy Receiving Report (ERR) in Wide Area Work Flow (WAWF).

(B) All additives line injected by the contractor shall be listed either on the CoQ, CoA, or equivalent, or on the shipping document. Either the concentration of the additive or the quantity of additive line injected into the aviation fuel is required on the documentation. Also annotate the additive brand name and ratio of fuel to additive if additives are cut with fuel prior to injection. Mixing additives together prior to injection is not allowed.

(2) The Contractor shall keep all quality and quantity records related to the performance of this contract complete and available to the Government during the performance of this contract and for three years after final payment under this contract.

(3) The contractor may provide transportation to/from/between contractor facilities and operations to the QAR/Government representative performing official duties relating to the administration of the contract and the contract price shall include the cost of any such transportation.

(b) The following requirements of this Quality Assurance Provision (QAP), paragraphs (b) (1) through (b) (7) and applicable tables, apply to all aviation fuel (not covered in paragraph (a) above), Marine Diesel (F76) and lubricating oil shipments, as well as all and Posts, Camps, and Stations (PC&S) non-aviation fuel shipments via pipeline or waterborne transport.

(1) QUALITY CONTROL PLAN (QCP)

(i) The Contractor is required to provide and maintain an inspection system, and a written description (QCP) of that system, acceptable to the Government. The Contractor has the option to provide and maintain an inspection system that, as a minimum, incorporates the requirements of ISO9001, Quality Management Systems. If the Contractor chooses to comply with the ISO9001 quality system requirements and format, all the requirements set forth in Quality Assurance Provisions (QAPs) contained in this contract shall be included in the resulting ISO9001 document that will serve as the QCP. A copy of the QCP, in English, shall be presented to the Quality Assurance Representative (QAR) for their review and acceptance prior to commencement of production or services. An acceptable QCP is required prior to Government inspection and acceptance of supplies or services. The QCP shall be reviewed annually and updated when deemed necessary by the Contractor or the Government, to include changes necessary to prevent the recurrence of quality problems. The annual review shall be performed by the Contractor and shall be documented in a manner that allows the QAR to validate the review date and who performed it. The Contractor shall sign and date the original QCP, as well as each subsequent revision. Revisions shall also be presented to the QAR for review and acceptance prior to implementation. When acceptable to the Government, the original QCP, and any revisions, shall be signed and dated by the QAR. The requirements outlined in the QCP do not change requirements of the contract; any changes to contract documents are executed by the Contracting Officer.

(ii) The Contractor shall require subcontractors to provide and maintain inspection/testing systems that adhere to all the requirements in this QAP.

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(iii) The QCP shall include contact information for key operational positions, a schematic diagram of facilities pertinent to the inspection system, indicating all inspection points. QCP shall include a description of the following operations related to the supplies or services to be furnished under the contract:

(A) Additive Injection and Blending Procedures used to ensure the quality of additives blended into the product(s) supplied under this contract. Procedures to identify and ensure the quality of component base stocks used to produce finished product. Procedures to be used for adding, prior to batching, all required additives at all locations. When procedures for in-line blending of non-aviation products in accordance with the IN-LINE BLENDING OF NON-AVIATION PETROLEUM PRODUCTS contract text (F109) are used, the QCP shall provide for establishing blend ratios, and identify the responsible personnel within the Contractor's organization authorized to establish the blend ratios. When line injection of additives is required by the contract, the QCP shall provide procedures for proportionately injecting additives throughout the entire loading process, to ensure homogeneous blending of all additives into the product, and procedures for maintaining records evidencing the blending of all line-injected additives. In addition to the testing identified in paragraph (b)(1)(iii)(C) of this QAP, when additives are to be line injected, a laboratory hand blend of aviation fuel or F76 with all additives required by the contract shall be tested to full specification prior to shipment to verify compliance. All additives line injected by the contractor shall be listed either on the Certificate of Quality (CoQ), Certificate of Analysis (CoA), or equivalent, or on the shipping document. Either the concentration of the additive or the quantity of additive line injected into the aviation fuel is required on the documentation. Also annotate the additive brand name and ratio of fuel to additive if additives are cut with fuel prior to injection. Mixing additives together prior to injection is not allowed.

(B) Sampling. Procedures for sampling additives, blend tanks, shipping tanks, lines, and conveyances/containers in accordance with American Petroleum Institute (API) Manual of Petroleum Measurement Standards (MPMS) 8.1/ASTM D 4057 - Manual Sampling of Petroleum and Petroleum Products, or API MPMS 8.2/ASTM D 4177- Automatic Sampling of Liquid Petroleum and Petroleum Products. For all tanker, barge, and pipeline shipments, a flow-proportional sample taken in accordance with ASTM D 4177, is preferred at the custody transfer point. However, manual samples taken in accordance with ASTM D 4057 are acceptable if they are taken hourly throughout the shipment and the quantity of the composite sample meets the requirements in Table I-Sample Retention of this QAP. Procedures shall include sample location, frequency, quantity, and retention.

(C) Testing. Types of test series and individual test methods/procedures to be performed on samples taken from each location identified below in Table II - Minimum Sampling and Testing Requirements for Petroleum Products and Table III - Definition of Test Series for tests required for the applicable product. For products not included in these tables, contact the QAR/Government representative for guidance. Full specification (A-level) testing shall be performed on each batch tank offered under the contract and results documented on a CoQ, CoA, or equivalent. Prior to performing A-level testing, tank sampling for homogeneity will be accomplished on new batches by obtaining upper, middle and lower samples. These samples shall be tested for API Gravity and the difference between any one of the samples shall be within 0.7 degrees API (for tanks composed of multiple batches) and 0.3 degrees API (for tanks composed of a single batch). If the difference between any of the samples are within these tolerances, the product is considered homogeneous and A-level testing shall be conducted. If the difference between any of the samples is greater than these tolerances, the following additional tests shall be conducted on product from the same the upper, middle and lower samples that were used to perform the API Gravity analysis: flash point, distillation initial boiling point and end point, density and freeze point.

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The results of these additional tests shall be forwarded to the QAR/government representative for approval and authorization to release the product for delivery. The Contractor shall perform all inspections and acceptance tests required by the specifications of the supplies to be furnished under this contract or, shall have such tests performed, in a laboratory acceptable to the Government. For FOB Destination contractor-supplied tankers/barges, the Contractor shall provide the QAR/Government representative, prior to the first shipment, with the name and location of the laboratory to be used for testing at each discharge point. The Contractor shall also notify the QAR/Government representative of any subsequent laboratory changes prior to the affected shipments. If any laboratory is determined to be unacceptable to the Government, the Contractor shall designate another laboratory that is acceptable, before acceptance testing can take place. Acceptance tests will normally be performed at the acceptance point. However, when such tests are performed at origin on supplies to be accepted at destination, documentation that will enable verification of the original test results shall be provided to the Government prior to acceptance at destination. Regardless of the method of delivery, products furnished under this contract shall meet all specification requirements at the custody transfer point (CTP).

(D) Calibration Program for testing and measuring equipment in accordance with ISO 10012, Measurement Management Systems - Requirements for Measurement Processes and Measuring Equipment, or equivalent local regulation, as appropriate. Program for meters used to determine quantity complying with the API MPMS, Chapters 4, 5, and 6, or equivalent foreign standard. For items not covered by API, ASTM International, or Energy Institute (EI) publications, the applicable manufacturer recommended calibration method(s), outlined in the applicable industry publication, shall be used if acceptable to the Government.

(E) Storage And Handling Procedures for quality/quantity determination. Includes a description of storage and handling equipment, such as tanks, lines, valves, and manifolds. Identification of dedicated/common product systems, including description of line segregation and controls to assure capability for proper gauging, sampling, draining of water, filtration, circulation, and drying. Identification of any other process/system used in maintaining product integrity during storage and handling.

(F) Loading and Shipping - General procedures for product movement and related quality/quantity checks from shipping tank(s) to CTP. Description of transfer system from shipping tank to CTP. A dedicated system, including lines, pumps, loading arms, and hoses, is preferred, but an isolated common system incorporating blind flanges, spectacle plates, or double valves between systems is acceptable. An isolated common system using single valves designed to ensure positive isolation, such as twin seal single valves, is also acceptable. Systems with single valve (other than twin seal) isolation require specific procedures be included in the QCP to ensure product integrity from the first single valve to the CTP. Procedures for conditioning and testing of isolated systems that last carried a product other than that intended for movement under this contract. For in-line blending of non-aviation products, where approved in the contract, requirements shall comply with the IN-LINE BLENDING OF NONAVIATION PETROLEUM PRODUCTS contract text/provision.

(G) Loading And Shipping - Tank Cars, Tank Trucks, And Intermodal/ISO Containers

For direct deliveries using Contractor-supplied tank cars and tank trucks in dedicated same grade aviation fuel service, refer to the section of API 1595 entitled “Loading of Road/Rail Cars” for loading and shipping procedures. All other tank car, tank truck, and intermodal/ISO container loadings shall be in accordance with the following procedures:

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(*1*) Confirm all compartments have been prepared in accordance with Table IV, Conversion Chart for tank trucks (TT), tanks cars (TC), tank wagons (TW), tank & trailer (TRR), and intermodal/ISo containers (IMC). Whenever possible, ensure the same preparation procedures used for compartments are applied to hoses, manifolds, and other handling equipment. If requested by QAR, provide documentation showing proof of previous cargo and cleaning procedures.

(*2*) Ensure that all conveyances are dry and free from loose rust, scale, and dirt.

(*3*) Provide for investigation of discrepancies in either recorded quality or quantity.

(*4*) When required by the contract, seal conveyance and record seal numbers on the shipping document.

(*5*) Strainers and filters shall be located as near the loading or filling point as practicable and shall be used as follows for all deliveries except deliveries into tanker, barge, or pipeline.

(*i*) All aviation fuel shall be passed through strainers of 60 mesh or finer screen;

(*ii*) All lubricating oil products, including preservatives, having a kinematic viscosity of 20.0 centistokes or less at 100°F shall be passed through a 100 mesh or finer screen;

(*iii*) All lubricating oil products, including preservatives, having a kinematic viscosity greater than 20.0 centistokes at 100°F, but less than 22.0 centistokes at 210°F, shall be passed through a 60 mesh or finer screen; and

(*iv*) The Contractor shall furnish and inspect strainers every 6-months in accordance with filter manufacture procedures , keeping a written record thereof.

(H) Loading and Shipping - FOB Destination (Contractor-supplied) Tankers and Barges. Procedures for the preparation of vessel cargo systems, in accordance with EI HM 50 - Guidelines for the Cleaning of Tanks and Lines for Marine Tank Vessels Carrying Petroleum and Refined Products, that ensure the vessel is suitable to load the intended product(s). Particular attention shall be given to vessels that previously carried a Fatty Acid Methyl Ester (FAME) cargo. As a minimum, FOB Destination Contractors shall ensure that all cargo systems on the vessel intended for loading product under this contract, to include tanks, lines, manifolds, and pumps, are washed, gas freed, and inspected. In the event a solvent degreaser is utilized in the tank preparation process, a minimum of two rinses shall be performed by the vessel prior to loading aviation fuel. These procedures, and others deemed necessary by the Contractor, shall be included to ensure that the product meets the specification requirements at the CTP.

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(I) Loading and Shipping - FOB Origin (Government-supplied) Tankers and Barges
Procedures for maintaining a time log of all significant events/delays including vessel notice of readiness, vessel arrival, mooring, vessel deballasting, inerting and conditioning of cargo tanks, inspections, hose connections and disconnections, starts, stops, vessel release, or any other event that affects laytime of the vessel. Procedures for assuring condition of loading line (full of tested product, all air bled, and pressure packed). Procedures for gauging shore tanks, both before and after loading. In the event an automated metering system is to be used as the means of shore quantity determination, manual gauges of all storage tanks designated for shipment shall be taken prior to commencement of loading. Procedures for Pre-Loading, to include procedures for discussion between Contractor (Supplier), vessel, and QAR/ Government representative, including, but not be limited to, prior three cargoes, cleaning procedures, loading plan, loading rates, sampling requirements, and after loading sampling and gauging. Pre-Load actions shall include: gauging, sampling, and C-level testing (for product verification purposes) of On-Board Quantity (OBQ) or Remaining On-Board (ROB) cargoes designated for load on top; Gauging and sampling (1 gallon) any other product on board, except for JPTS, and retaining those samples; All cargo quantities shall be calculated and volume corrected to 60°F or 15°C both before and after loading. Procedures for commencement of loading into one tank (up to 3 feet) or line displacement, whichever is greater, then switching to at most two other vessel tanks while first-in sampling and testing is being accomplished (See Table II). Procedures for the transportation of samples from the vessel to the testing facility. Procedures for monitoring the loading (including line blending and additive injection) from source to vessel, investigating irregularities immediately and stopping loading, if necessary. Procedures for investigating discrepancies in quality (mandated if off-specification or out of testing tolerance) and quantity (mandated if shore to ship variance is in excess of the following: 0.2% (0.002) for cargos not requiring cleaning, gas freeing, or drop/strip; 0.3% (0.003) for cargos requiring drop/strip; or 0.5% (0.005) for cargos requiring cleaning and gas freeing) on loaded conveyance.

(J) Loading and Shipping Tankers and Barges - General (FOB Origin and Destination). Procedures for immediately notifying the QAR/ Government representative when irregularities occur or are suspected, and on all occasions when loading is interrupted. *Authority to release a government-furnished vessel rests with the QAR/Government representative after compliance and completion by the Contractor of all required operations, including the preparation of the DD Form 250-1 or Wide Area Workflow (WAWF) Energy Receiving Report (ERR) with Time Statement.*

(K) Records And Reports. Procedures for completing and distributing required documentation both prior to and after the release of the shipping conveyance. Documentation shall include, as a minimum, test reports for all products and additives, additive blending and/or injection records, vessel notice of readiness, customs documents, bills of lading, vessel ullage reports, and the DD Form 250/DD Form 250-1 and continuation sheet(s) (or the WAWF ERR documents). Other documentation may be required on an as needed basis, such as vessel port logs, calibration records, and results of quality or quantity discrepancy investigations. These records and reports shall include by whom, where, and how they were prepared, as well as retention information. The DD Form 250-1 and DD Form 250-1 continuation sheet(s), or WAWF ERR with Time Statement shall be signed by the Contractor and the vessel representative in the appropriate blocks before presenting them to the QAR/Government representative for signature. The DD Form 250 and DD Form 250-1 or WAWF ERR documents shall identify the type, brand name, ratios of additives cut with fuel, and amount of all additives injected. A copy of the CoQ, CoA, or equivalent, shall be provided to the QAR/Government representative for approval prior to shipment. CoQ/CoA results shall be entered by the contractor into the Statement of Quality Tab during submission of an ERR in WAWF. A copy of the CoQ, CoA, or equivalent, and Bill of Lading shall also be provided to the truck driver for all tank truck/trailer or intermodal shipments and the railroad representative for all rail car shipments.

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(L) Corrective Action. Procedures shall address correction of any deficiency affecting product quality or quantity. The corrective action procedures shall include notification of the QAR/Government representative.

(2) Prior to loading, the Contractor shall inspect all shipping conveyances for suitability to load the supplies furnished under this contract, except for Government-furnished tankers and barges, which shall be inspected by the QAR/Government representative. Authorization for the Contractor to participate in the inspection of Government-supplied tankers and barges rests with QAR/Government representative. The QAR/ Government representative has the right to inspect any Government-supplied conveyance prior to loading and, in the event the Contractor and the QAR/Government representative disagree on the suitability to load such a conveyance, the determination of the QAR/Government representative shall govern. If the SHIPMENT AND ROUTING contract text/provision is included in the contract, Government-furnished transportation equipment that is unsatisfactory for loading shall be reported by the Contractor in accordance with the terms contained in that contract text/ provision. Procedures to determine suitability to load tank trucks and tank cars shall include, but not be limited to, visual inspection of interior compartments to assure cleanliness and dryness. Manifolds and hoses shall be drained and be clean and dry for the intended product. If the Contractor intends to witness operations at any point up to, and including, government acceptance, the Contractor is responsible for contacting the facilities to be visited and obtaining the required access.

(3) When requested by the Government, the Contractor shall provide samples of each product furnished under this contract, in the manner and to the place designated by QAR. Such samples shall be provided without charge to the Government and shall be packed, marked, and shipped by the Contractor, at the Contractor's expense, in sampling containers and shipping boxes provided by the Contractor. During this contract, the Contractor shall be required to provide no more than five (ten in the case of aviation turbine fuel and F-76) 1-gallon samples under each contract line item (CLIN). The Contractor may be required to provide samples larger than 1-gallon, but in no case shall the total quantity for all samples under each CLIN exceed the quantities stated in this paragraph. The samples in this paragraph are in addition to any samples required elsewhere in this contract.

(4) The Contractor shall keep all quality and quantity records related to the performance of this contract complete and available to the Government during the performance of this contract and for three years after final payment under this contract. These records shall include, but not be limited to, DD Form 250 series or WAWF ERR series documents, documentation of internal inspections of all conveyances, CoQ/CoA's for all required testing (including those for the shipping tanks for FOB Destination tankers/barges), and documentation of quantity determinations.

(5) When a common system is to be used for shipment in accordance with paragraph (b)(1)(iii)(F) of this QAP, the Contractor shall provide a list of all products carried by that system during the past year to the Contracting Officer, at the time of offer, for a determination of acceptability.

(6) The inspection system and related operations provided or performed pursuant to this QAP shall be subject to surveillance by the QAR/ Government representative.

(7) The contractor may provide transportation to/from/between contractor facilities and operations to the QAR/Government representative performing official duties relating to the administration of the contract and the contract price shall include the cost of any such transportation.

TABLE I, Sample Retention[1]

LOCATION	MINIMUM QUANTITY	RETENTION PERIOD [1]
Bulk Tank Cars/Tank Truck/Intermodal Container		
Fuels	1 gallon (4 liters) [2]	15 days for fuels
Lubricating Oils	0.25 gallons (1 liter) [2]	45 days for lubricating oils
Drums[4]	2 gallons (8 liter) (multi-drum composite) [3]	45 days
Shipping Tank(s)	2 gallons (8 liters) for Aviation Turbine Fuels 1 gallon (4 liters) for all other Fuels, Lubricating Oils, and Additives	45 days
Tanker/Barge/Pipeline (Composite Line at Custody Transfer Point)	5 gallon (20 liters) for all fuel types	45 days
Tanker/Barge (Composite)		
F.O.B. Origin	2 gallons (8 liters) for Aviation Turbine Fuels 1 gallon (4 liters) for all other fuels	45 days
F.O.B. Destination	2 gallons (8 liters)	Until shore tests on-specification
Tanker/Barge (Each Compartment)		
F.O.B. Origin	0.25 gallon (1 liter)	45 days
F.O.B. Destination	0.25 gallon (1 liter)	Until shore test on-specification

NOTES:

1. Unless otherwise noted in this document or applicable contract, the retention period shall be for the stated number of days after the sample has been taken from the location indicated.
2. If the conveyance is compartmentalized, a proportional sample shall be taken from each compartment.
3. An 8-liter composite sample of each product lot shall be taken from a predetermined number of randomly selected drums IAW Table III of API MPSM, Chapter 8.1 (ASTM D 4057).
4. Applies to DLA Energy managed procurements of products shipped in drums. Does not apply to DLA Aviation managed procurements for packaged products.

TABLE II, Minimum Sampling and Testing Requirements for Petroleum Products
Series 1, tankers and barges[1]

ITEM	WHEN TO SAMPLE		LOCATION OF SAMPLE	TYPE OF SAMPLE[2]	TEST REQUIRED[3]
1.a	Prior to Loading	Before acceptance of new materiel and after establishment of new batch[4]	Each shore storage tank	Single Tank Composite or All Levels	A[5, 15]
1.b		After line pack but prior to loading	As close to the custody transfer point as possible	Line Sample	C[7,14] Dedicated System B-1 Non-Dedicated System
1.c	During Loading	Immediately after start of loading/shipment	As close to the custody transfer point as possible	Line Sample	C[7]
1.d		After first in	Vessel cargo compartment(s)	Spot	C[7,14]
1.e		During loading	As close to the custody transfer point as possible	Line Sample	Hourly Visual[6,8], Retain[10]
1.f	After Loading	After loading	Each vessel cargo compartment	All Level (Combine All Levels into one Volumetric Composite Sample[13])	C [6, 8, 11,14] Remainder of B-1[12] (Retain fuel for B-1 referee testing)

NOTES:

1. Additional requirements to include retain samples may be required. See Quality Assurance Provisions (C or E-QAP) for further details.
2. Refer to ASTM International D4057 for sampling procedures.
3. Test results shall be compared against the quality release certificate. Any discrepancies shall be identified, investigated, and resolved.
4. Upon procurement at refineries, blending installations, etc. and at main installations including national depots & ocean importing points on establishment of new batches.
5. Prior to performing A-level testing, the homogeneity of each batch shall be determined. Refer to paragraph (b)(1)(iii)(C) of this document for additional details.
6. Test for FSII and SDA only if line injected.
7. If sample fails appearance, perform particulate matter using approved methods as outlined in the applicable specification.
8. Lubricity also required for F-76 only if line injected, or determine LIA concentration by volumetric measurement/calculation.
9. First in shall be done after one (1) meter or after line displacement whichever is the greater.
10. For all tanker and barge, a flow-proportional sample taken in accordance with MPMS Chapter 8, Section 2, is preferred at the custody transfer point. However, manual samples taken in accordance with MPMS Chapter 8, Section 1 are acceptable if they are taken hourly throughout the shipment and that the quantity of the composite sample meets the requirements in Table I
11. Best practice is for B-1 test completion for FOB Origin Procurements (or Type A level testing when specified in contract for FOB Destination Procurements) prior to vessel release. Based on QAR decision, vessel may sail after C Tests to meet schedule requirements including to avoid predicted weather; remainder of B-1 tests to be completed before arrival at next load or discharge port. Test report will be provided to designated discharge port.
12. Type A level testing required for FOB Destination Procurements (unless otherwise specified by contract).
13. As defined by API MPMS Chapter 8-1, section 3.22.
14. For product verification purposes. Results of C-test must fall within reproducibility of source tank A-level results.
15. New cargo from same batch, within 30 days, will only require B-1 testing to verify previous A-level test results. B-1 level verification test results should fall withing reproducibility of source tank A-level results.

TABLE II, Minimum Sampling and Testing Requirements for Petroleum Products
Series 2, pipeline, tank trucks, tank cars, iso-containers[1,7]

ITEM	WHEN TO SAMPLE		LOCATION OF SAMPLE	TYPE OF SAMPLE [2]	TEST REQUIRED[3]
2.a	Prior to Shipment	Before acceptance of new materiel and after establishment of new batch[4]	Each Shipment Storage Tank	Single Tank Composite or All Levels	A[9, 11]
2.b		After line pack but prior to loading/shipment	As close to the custody transfer point as possible	Line Sample	C[5,6] Dedicated System B-1 Non-Dedicated System
2.c	During Shipment	Immediately after start of loading/shipment	As close to the custody transfer point as possible and downstream of the filter separator (if equipped)	Line Sample	C[5, 13]
2.d		Pipeline Only: Hourly after start of loading or shipment[10]	As close to the custody transfer point as possible and downstream of the filter separator (if equipped)	Line Sample	Apperance[5, 12] (Retain fuel for B-1 referee testing)
2.e		Daily on first fill and anytime a new tank is placed on issue (after line displacement) TT, TC, ISO, Refuelers or other over the road transportation modes only	As close to the custody transfer point as possible (Loading Rack or Fill Point) and downstream of the filter separator (if equipped)	Line Sample	Color (by visual method) and Appearance[5]. If visual fails perform C[6]
2.f	After Loading	Daily for each conveyance mode (Not required for pipeline operations)	Each Cargo Compartment or another point that provides a representative sample	All Level	Appearance on each compartment C on composite [5,6,8]
2.g		Drums	Sample of each lot of product shall be taken from a predetermined number of randomly selected drums[14]	Multi-drum Composite	C[13]

NOTES:
1. Additional contractual requirements to include retain samples may be required. See Contract Quality Assurance Plans (C or E-QAPs) for further details.
2. Refer to ASTM International D4057 for sampling procedures.
3. Test results shall be compared against the quality release certificate. Any discrepancies shall be identified, investigated, and resolved.
4. Upon procurement at refineries, blending installations, etc., and at main installations including national depots & ocean importing points on establishment of new batches.
5. Test for FSII and SDA only if line injected.
6. If sample fails appearance, perform particulate matter using approved methods as outlined in the applicable specification.
7. Type A level testing required for FOB Destination Procurements (unless otherwise specified by contract).
8. If the conveyance is compartmentalized, a proportional sample shall be taken from each compartment such that a total of one liter is collected from the conveyance.
9. Filtration time testing is required for JP-5 & JP-8 and shall meet contract requirement.
10. For pipeline shipments, a flow-proportional sample taken in accordance with MPMS Chapter 8, Section 2, is preferred at the custody transfer point. However, manual samples taken in accordance with MPMS Chapter 8, Section 1 are acceptable if they are taken hourly throughout the shipment and that the quantity of the composite sample meets the requirements in Table I.
11. Prior to performing A-level testing, the homogeneity of each batch shall be determined. Refer to paragraph (b)(1)(iii)(C) of this document for additional details.
12. Lubricity also required for F-76 only if line injected or determine LIA concentration by volumetric measurement/calculation.
13. For product verification purposes. Results of C-test must fall within reproducibility of source tank results.
14. A composite sample of 2 gallons (8 liters) for Aviation Turbine Fuels and 1 gallon (4 liters) for all other fuels for each lot of product shall be taken from a predetermined number of randomly selected drums. The total number of drums sampled per lot shall, at a minimum, be that cited in Table III of API MPMS chapter 8.1 (ASTM d 4057), but may be increased at the government's option.

TABLE II, Minimum Sampling and Testing Requirements for Petroleum Products
Series 3, product receipt and discharges for tankers and barges (FOB Destination)

ITEM	WHEN TO SAMPLE		LOCATION OF SAMPLE	TYPE OF SAMPLE	TEST REQUIRED
3.a	Prior to Discharge	Prior to discharge	From each vessel cargo compartment	All Level	Appearance and Density[1]
3.b			From each vessel cargo compartment	All Level (Combine All Levels into one Volumetric Composite Sample)	A
3.d	During Discharge	Within 30 minutes after start and each hour after until completion	Dock/Discharge Manifold Header	Line Sample / Retain	Perform Appearance test[2,3] and then retain fuel for Referee testing
3.e	After Discharge	If remainder of fuel will be shipped to another DFSP	Each vessel compartment with fuel remaining on board	All Level	Retain until discharged at destination

NOTES:
1. Test for flash point on high flash cargo tanks if low and high flash cargoes are carried together aboard the same vessel without segregated venting systems.
2. For aviation fuel delivery, FSII and SDA shall be tested only if line injected. This requires an additional quart sample.
3. Test for additives if line injected downstream of the injector. (If line injecting LIA into F-76, test for lubricity.)

TABLE II, Minimum Sampling and Testing Requirements for Petroleum Products
Series 4, product receipt from pipeline (FOB Destination)[1]

ITEM	WHEN TO SAMPLE		LOCATION OF SAMPLE	TYPE OF SAMPLE	TEST REQUIRED
4.c	During Receipt	During pipeline receipt	Custody transfer point	Line Sample/Retain	5 Gallon Minimum[2]
4.d	During Receipt	During pipeline receipt, immediately after start, Midpoint and 1 hour prior to completion[3]	As close to the custody transfer point as possible	Line	C
4.g	After Receipt	After receipt of fuel	Each receipt tank	All level	A

NOTES:
1. Refer to ASTM International D4057 for sampling procedures.
2. Retain for referee testing in the event a failure occurs on receipt tank. Retain sample for 45 days. Sampling is IAW API MPMS Chapter 8
3. Fuel from a new batch will be considered a new receipt for quality control purposes.

TABLE III, Definitions of Test Series

I. TYPE A: Includes all specification quality conformance tests plus any additional contractual requirements.

II. TYPE B-1 & C: As shown in the table below for each product. Properties and test methods will be in accordance with the product specification for each grade identified in the solicitation/contract.

	AVGAS		AVIATION TURBINE		MOGAS		DIESEL/ KEROSENE/ NAVAL DISTILLATE		LUBE OIL	
TEST PROPERTIES	**B-1**	**C**	**B-1**	**C**	**B-1**	**C**	**B-1**	**C**	**B-1**	**C**
Appearance/ workmanship	*	*	*	*	*	*	*	*	*	*
Particulate content			*		*[8]		*[1]		*[8]	
Filtration Time			*							
Color (visual)	*	*	*	*	*	*	*	*	*	
Density/API Gravity/ Specific Gravity	*	*	*	*	*	*	*	*	*	*
Distillation	*		*		*		*			
Corrosion, Copper Strip	*		*							
Existent Gum	*		*							
Carbon Residue							*[1]			
Vapor Pressure	*				*					
Lead Content	*		*[3]							
Freeze Point			*							
Flash Point			*	*			*	*	*	
FSII Content			*							
MSEP (JP-8, F-24, F-27, Jet A & Jet A-1)			*							
Conductivity (JP-5 & JP-8, F-24, F-27 & Jet A-1)			*[4]							
Thermal stability (B-1 requirement applies to JPTS and marine/multi-product mode receipt tank sample)			*							
Sediment & Water										
Total Sediment aged										
Viscosity									*[7]	
Water Content	*				*[8]				*[2]	
Emulsion									*[2]	
Foam Test									*	
Knock Rating (RON / MON)					*[6]					
Lubricity							*[1]			
FAME			*[5]				*[1]			
Solid Contaminants									*[2]	

** The procedure to be used for conducting these tests will be as stated in the appropriate product specification and/or contract. For any products not covered under this table, contact Quality Assurance Representative for guidance.*

NOTES:

1. F76 only.
2. Only if the specification has this requirement.
3. If contamination with leaded fuels is suspected.
4. If fuel contains conductivity additive, wait approximately 2 minutes after obtaining sample for charges in the fuel to dissipate before performing CU testing.
5. Only for product transported through multi-product systems, including marine transport.
6. Perform only if Lead Content is performed and fails and/or contamination with another product is suspected. In the case of pipeline, this test shall be performed when considered necessary.
7. For MIL-PRF-17331 and MIL-PRF-9000 viscosity is not required unless the tank has been in long term storage for 3 months.
8. Visual check.

TABLE IV, Conversion chart for tank trucks (TT) tank cars (TC), tank wagons (TW), tank & trailer (TRR), and intermodal containers (IMC)[1, 2]

Last Product Carried	PRODUCT TO BE LOADED[5]								
	Gasolines: MUR, MUM, MUP, EFM, E85, 100, 130, JAB	Aviation Turbine Fuels: JAA/A1, JP-8, F-24, JP-5, TS1, KS1/R/N,	Aviation Turbine Fuel[3]: JPTS	RP1/RP2	JP-10/PF1	Diesel Fuel: F-76	FSII	Lubricating Oils	Diesel Fuels: DF1, DF2, DS1, DS2, DSS, DSW, BDI, MGO
Gasolines[4] MUR, MUM, MUP, EFM, E85, 100, 130, JAB	Drain/Empty	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry
Aviation Turbine Fuels[4, 9] JAA/A1, JP-8, F-24, JP-5, TS1, KS1/R/N,	Drain/Empty	Drain/Empty	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry
Aviation Turbine Fuel[3] JPTS	Drain/Empty	Drain/Empty	Drain/Empty	Drain/Empty	Steam/Dry	Steam/Dry	Steam/Dry	Steam Dry	Steam/Dry
RP1/RP2	Steam/Dry	Steam/Dry	Steam/Dry	Drain/Empty	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry
J10/PF-1	Steam Dry	Steam/Dry	Steam/Dry	Steam Dry	Drain/Empty	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry
Petroleum Solvent or Paint Thinner	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry
Diesel Fuel: F-76	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry	Drain/Empty	Steam/Dry	Steam/Dry	Drain/Empty
Diesel Fuels[4] DF1, DF2, DS1, DS2, DSS, DSW, MGO	Steam/Dry	Steam/Dry[7]	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry	Drain/Empty
FSII (SIH)	Drain/Empty	Drain/Empty	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry	Drain/Empty	Steam/Dry	Steam/Dry
Lubricating Oils	NO LOAD	NO LOAD	NO LOAD	NO LOAD	NO LOAD	NO LOAD	NO LOAD	Steam/Dry[6]	Steam/Dry
Biodiesel Blends($\leq$B15)[4] DF1, DF2, DS1, DS2, DSS, DSW	Steam/Dry	Steam/Dry[7]	NO LOAD	NO LOAD	NO LOAD	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry
Biodiesel Blends(>B15): DF1, DF2, DS1, DS2, BDI	NO LOAD	NO LOAD	NO LOAD	NO LOAD	NO LOAD	NO LOAD	NO LOAD	NO LOAD	Steam/Dry[8]
Heavy Fuel Oils, Distillate Fuels[4] FS1, FS2, FS4, FS5, FS6 Residual Fuel Oils: ISO 8217, class F (IFOs)	NO LOAD	NO LOAD	NO LOAD	NO LOAD	NO LOAD	NO LOAD	NO LOAD	NO LOAD	Steam/Dry
Naphtha	Drain/Empty	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry	Steam/Dry

TABLE IV Continued, Conversion chart for tank trucks (TT) tank cars (TC), tank wagons (TW), tank & trailer (TRR), and intermodal containers (IMC)[1, 2]

NOTES:

1. For previous cargoes not listed, contact DLA Energy Quality Operations Division at DLAEnergyQualityOperations@dla.mil for acceptability/cleaning procedures.
2. General Instructions for Tank Cars and Trucks
   a. Conveyances (all) shall be dry and free from loose rust, scale, and dirt
   b. Saran lined equipment should not be steam cleaned; water wash should suffice.
   c. Petroleum products shall not be loaded into the conveyances where the previous cargo was caustic, acid, or chlorinated solvents.
   d. Tank trucks and cars in liquid fertilizer service shall not load aviation turbine fuels directly but shall carry out at least two loads of commercial gasoline prior to the aviation turbine fuel load.
   e. Conversion of Government-owned tank cars from liquid fertilizer service to aviation turbine fuel service shall only be done when no other alternative exists. Tank cars being converted from liquid fertilizer service to a petroleum product shall be adequately cleaned to remove all traces of liquid fertilizer. At a minimum, the equipment must be steam cleaned; dried and must be free from loose rust, scale, and dirt. After cleaning, but before loading aviation turbine fuels, at least two loads of commercial gasoline shall be carried prior to loading aviation turbine fuel. Equipment with unlined compartments should have its compartments lined with an approved coating at this time. Conveyances carrying their initial load of aviation turbine fuel following conversion shall not be released from the loading until they have been sampled a minimum of 24 hours after loading. In accordance with MIL-STD-3004-1A, samples shall undergo type B-2 testing plus Thermal Stability Test (if required by the product specification). After type B-2 tests indicate compliance with specification requirements, the conveyance can be released pending results of the Thermal Stability tests. The conveyance shall not be unloaded until the loading point notifies the destination that the Thermal Stability test (when required) has passed. When the above condition applies, the DD-250, Wide Area Workflow (WAWF) Energy Receiving Report (ERR) or equivalent form shall be so noted.
3. To be loaded only in aluminum or stainless-steel conveyance, or other conveyance lined with an approved epoxy coating. If equipment is coated, clean with hot fresh water not exceeding 58°C (136°F) and dry thoroughly.
4. If previous cargo contained dye marker, all traces of color must be removed.
5. If product to be loaded does not contain dye, then the vehicle shall not contain any traces of dye prior to loading.
6. Drain/empty only when same specification product is being loaded.
7. Prior to loading the aviation fuels listed, the conveyance must have carried a previous non-FAME cargo and then be steam cleaned and dried. Trucks that have previously carried biodiesel create a risk of cross-contamination of aviation turbine fuel with residual biodiesel components. Diesel fuel may contain up to 5% FAME without reporting it as biodiesel. Because of the difficulties in making a load-rack of the presence of FAME in the previous load, the more stringent "Steam/Dry is required.
8. Drain and empty if loading biodiesel blend.
9. Cargos carrying dyed kerosene shall use the steam/dry cleaning procedure.